

About Me

I am a Ph.D. candidate in Computer Science at Georgia Tech, advised by Prof. Vijay Ganesh and Prof. Sriram Vishwanath. I spent one year of my Ph.D. at the University of Waterloo, Canada. My research lies at the intersection of automated reasoning and machine learning: I combine symbolic reasoning and formal methods with large language models (LLMs) through reinforcement learning, semantic alignment, and other techniques to significantly advance their capabilities in software engineering (code translation, generation, and repair) and mathematical reasoning (proof auto-formalization, synthesis, and discovery in Lean). My work has appeared at ICSE, ICLR, IJCAI, ECAI, WWW, and received a Distinguished Paper Award at ICSE 2026.

Previously, I earned my Master's degree in Computer Science from IIT Kharagpur (Institute Silver Medalist, 2022) and my Bachelor's degree in Computer Science from Jadavpur University (University Gold Medalist, 2020), graduating at the top of my class at both institutions. In Summer 2025, I interned at Amazon Science (AWS, Seattle) as an Applied Scientist with the Next Gen DevX team, designing LLM-based coding agents. I am also a recipient of the Reliance Foundation Scholarship for AI & CS (2021), and prestigious national awards JBNSTS, KVPY, and NTSE from the Government of India.

Research Areas: NEURO-SYMBOLIC AI · PROGRAMMING LANG. & COMPILERS · FORMAL METHODS · AI FOR SE · AI FOR MATH · LLM

Technical Skills

- **Languages:** Python, C, C++, Java, Lean, MATLAB, JavaScript
- **Software Tools:** PyTorch, Huggingface, TRL, DeepSpeed, Scikit-learn, NLTK, lex/flex, yacc/bison, LLVM-IR, Git, $\LaTeX$
- **Operating Systems:** Linux, MacOS, Windows

Languages

- **English:** Full Professional Working Proficiency
- **Hindi:** Full Professional Working Proficiency
- **Bengali:** Native or Bilingual Proficiency

Education

Georgia Institute of Technology (Georgia Tech), ATLANTA, USA

Ph.D. in Computer Science

– Advisor: Profs. Vijay Ganesh & Sriram Vishwanath

Aug 2023–Present

GPA: 4.0/4.0

University of Waterloo, ONTARIO, CANADA

Ph.D. in Computer Science (Transferred)

– Advisor: Prof. Vijay Ganesh

2022–2023

Grade: 90.33%

Indian Institute of Technology (IIT) Kharagpur, INDIA

M.Tech in Computer Science & Engineering

- ★ **Institute Silver Medalist**, Department Rank: 1st/75
- Thesis: “Fairness in Virtual Conference Scheduling” [PDF] (*Best Master's Thesis Nomination*)
- Advisor: Prof. Niloy Ganguly

2020–2022

CGPA: 9.84/10

Jadavpur University, KOLKATA, INDIA

B.E. in Computer Science & Engineering

- ★ **University Gold Medalist**, Department Rank: 1st/70
- Thesis: “A Real-Time Semantic Image Segmentation Approach Through Region-Partitioning” [PDF]
- Advisor: Prof. Sanjoy K. Saha

2016–2020

CGPA: 9.57/10

Work Experience

Amazon Web Services (AWS), SEATTLE, USA

Applied Scientist II Intern · Next Gen DevX (NGDE) Team, Amazon Science

- Manager: Bhavana Bhasker · Mentor: Sam Davidson
- Led development of a neuro-symbolic coding agent for Infrastructure-as-Code; received the **highest performance rating** (*Exceeded Expectations*) and a **nomination** for Georgia Tech's *Intern of the Year*.
- Published a **first-author paper** at **ICSE 2026** (CORE A* conf.), earning a **Distinguished Paper Award**.

May 2025 – Aug 2025

Publications

🔗 : [Google Scholar](#) (As of Apr 2026: citations = 234, h-index = 7, i10-index = 5)

◆: A or A* venues (prestigious and highly-selective) per the CORE Computer Science Conference Rankings

PEER-REVIEWED INTERNATIONAL CONFERENCE PAPERS

- [13] **ProofBridge: Auto-Formalization of Natural Language Proofs in Lean via Joint Embeddings** ◆ ICLR 2026
P. Jana, K. Kale, A. E. Tanriverdi, C. Song, S. Vishwanath, and V. Ganesh
14th International Conference on Learning Representations, Rio de Janeiro, Brazil, 2026. (Acpt. Rate: 28%) [[arXiv](#)]
- [12] **TerraFormer: Automated IaC with LLMs Fine-Tuned via Policy-Guided Verifier Feedback** ◆ ICSE 2026
P. Jana, S. Davidson, B. Bhasker, A. Kan, A. Deoras, and L. Callot
IEEE/ACM 48th Int'l Conf. on Software Engineering, SEIP Track, Rio de Janeiro, Brazil, 2026. [[Link](#)] [[arXiv](#)] [[YouTube](#)]
(★ Distinguished Paper Award)
- [11] **RLSF: Fine-tuning LLMs via Symbolic Feedback** ◆ ECAI 2025
P. Jha, P. Jana, P. Suresh, A. Arora, and V. Ganesh
28th European Conf. on Artificial Intelligence, Bologna, Italy, pp. 1687–94, 2025. (Acpt. Rate: 23%) [[Link](#)] [[arXiv](#)]
- [10] **CoTran: LLM-based Code Translator using RL with F/b from Compiler and Symbolic Exec.** ◆ ECAI 2024
P. Jana, P. Jha, H. Ju, G. Kishore, A. Mahajan, and V. Ganesh
27th European Conf. on Artificial Intelligence, Santiago, Spain, pp. 4011–18, 2024. (Acpt. Rate: 23%) [[Link](#)] [[arXiv](#)]
- [9] **NeuroSymbolic LLM for Mathematical Reasoning and Software Engineering** ◆ IJCAI 2024
P. Jana
33rd Int'l Joint Conf. on Artificial Intelligence, Doctoral Consortium, pp. 8492–93, 2024. (Acpt. Rate: 17%) [[Link](#)]
- [8] **Scheduling Virtual Confs. Fairly: Achieving Equitable Participant and Speaker Satisfaction** ◆ WWW 2022
G. K. Patro, P. Jana, A. Chakraborty, K. P. Gummadi, and N. Ganguly
31st ACM Web Conference, Lyon, France, pp. 2646–2656, 2022. (Acpt. Rate: 17.7%) [[Link](#)] [[PDF](#)]
- [7] **Unsupervised Action Localization Crop in Video Retargeting for 3D ConvNets** TENCN 2021
P. Jana, S. Bhaumik, and P. P. Mohanta
2021 IEEE Region 10 (Asia-Pacific) Conference, Auckland, New Zealand, pp. 670–675, 2021. [[Link](#)] [[PDF](#)]
- [6] **AutoDrone: Shortest Optimized Obstacle-Free Path Planning for Autonomous Drones** ICDC 2021
P. Jana and D. Jana
International Conf. on Computational Intelligence, Data Science & Cloud Computing, pp. 1–13, 2021. [[Link](#)] [[PDF](#)]
- [5] **Two Stage Semantic Segmentation by SEEDS & Fork Net** CALCON 2020
A. Mukherjee, P. Jana, S. Chakraborty, and S. K. Saha
2020 IEEE Calcutta Conference, pp. 283–287, 2020. [[Link](#)] [[PDF](#)]
(★ Best Student Paper 2nd Runner-Up Award)
- [4] **A Multi-tier Fusion Strategy for Event Classification in Unconstrained Videos** PReMI 2019
P. Jana, S. Bhaumik, and P. P. Mohanta
8th International Conf. on Pattern Recognition & Machine Intelligence, Springer, pp. 515–524, 2019. [[Link](#)] [[PDF](#)]
- [3] **Key-Frame based Event Recognition in Unconstrained Videos using Temporal Features** TENSYP 2019
P. Jana, S. Bhaumik, and P. P. Mohanta
2019 IEEE Region 10 (Asia-Pacific) Symposium, pp. 349–354, 2019. [[Link](#)] [[PDF](#)]
- [2] **Handwritten Document Image Binarization: An Adaptive K-means based Approach** CALCON 2017
P. Jana, S. Ghosh, S. K. Bera, and R. Sarkar
2017 IEEE Calcutta Conference, pp. 226–230, 2017. [[Link](#)] [[PDF](#)]
- [1] **A Fuzzy C-Means Based Approach Towards Efficient Document Image Binarization** ICAPR 2017
P. Jana, S. Ghosh, R. Sarkar, and M. Nasipuri
9th International Conference on Advances in Pattern Recognition (ICAPR), pp. 332–337, 2017. [[Link](#)] [[PDF](#)]

- [3] **SALiEnSeA: Spatial Action Localization & Temporal Attention for Video Event Recognition** IJCISIM 2022
P. Jana, S. Bhaumik, and P. P. Mohanta
Int'l Journal of Computer Info. Systems and Industrial Mgmt. Appl., vol. 14, pp. 270–284, MIR Labs, 2022. [Link] [PDF]
- [2] **Recent Trends in 2D Object Detection and Applications in Video Event Recognition** River 2022
P. Jana and P. P. Mohanta
Advances of Deep Learning and its Appl. in Object Detection and Recognition, River Publishers, 2022. [Link] [PDF]
- [1] **Event and Activity Recognition in Video Surveillance for Cyber-Physical Systems** Springer 2021
S. Bhaumik, P. Jana, and P. P. Mohanta
Emergence of Cyber Physical System & IoT in Smart Automation and Robotics, Springer Nature, 2021. [Link] [PDF]

PREPRINTS / UNDER SUBMISSION

- [2] **Abstractions-of-Thought: Intermediate Repr. for LLM Reasoning in Hardware Design** arXiv 2025
M. DeLorenzo, K. Tieu, P. Jana, P. Jha, D. Kalathil, V. Ganesh, and J. Rajendran
arXiv preprint arXiv:2505.15873, 2025. [arXiv]
- [1] **Reducing Opinion Echo-Chambers by Intelligent Placement of Moderate-Minded Agents** arXiv 2023
P. Jana, R. R. Choudhury, and N. Ganguly
arXiv preprint arXiv:2304.10745, 2023. [arXiv]

Scholarships and Awards

DURING PH.D.

- **Distinguished Paper Award, ICSE-2026** 2026
One of 3 papers recognized among 74 accepted in the SEIP track (IEEE/ACM 48th Int'l Conf. on Software Engineering)
- **2nd Place, College of Computing (CoC) Graduate Poster Symposium, Georgia Tech** 2025
Runner-up at the college-wide graduate research symposium across SCS, SCP, IC, CSE; evaluated by faculty
- **Top Reviewer, NeurIPS-2024** 2024
Top 8% of reviewers (1,300 out of 15,160); awarded complimentary full conference registration
- **Lunch with a EurAI Fellow Invitee, ECAI-2024** 2024
Invited by ECAI-2024 for a select Ph.D. lunch discussion with Prof. Mehdi Dastani (EurAI Fellow; Chair, ECAI-2025)
- **IJCAI-2024 Travel Grant** 2024
Travel grant to attend the 33rd International Joint Conference on AI; Jeju Island, South Korea
- **CAV-2024 Travel Scholarship** 2024
Full scholarship to attend the 36th International Conference on Computer-Aided Verification; Montréal, Canada

DURING MASTER'S

- **Institute Silver Medalist, CSE Department Topper** 2022
Highest GPA in M.Tech. Computer Science, IIT Kharagpur; Rank 1st/75
- **Best M.Tech. Thesis Award Nominee, CSE Department** 2022
Nominated for Best Thesis, M.Tech. Computer Science, IIT Kharagpur
- **Reliance Foundation Scholarship for AI & Computer Science** 2021
One of 38 Master's students selected across India; INR 0.6 million in academic funding
- **CSI Young IT Professional Award – East Region Runner-Up** 2021
Runner-up in the Eastern India Region; youngest participating team in the competition
- **GATE Scholar (Graduate Aptitude Test in Engineering)** 2020
All-India Rank 188/100,000+ (99.81 percentile); stipend of INR 12,400/month for 2 years

DURING BACHELOR'S & HIGH-SCHOOL

- **University Gold Medalist, CSE Department Topper** 2020
Highest GPA in B.E. Computer Science, Jadavpur University; Rank 1st/70

• Top 6 in TIFR GS-2020 (Computer Science) Nationwide Among 6 students selected across India for direct Ph.D. admission at TIFR, India via competitive exam and interview	2020
• CSE Department Annual Gold Medalist Highest annual GPA in CSE, Jadavpur University, for three consecutive years	2017–2019
• Summer Research Fellowship Programme (SRFP) Selected by the Indian Science Academies among the top 350 undergrads nationwide; two consecutive years	2018, 2019
• Special Mention Award, Computer Vision Summer School (SSCVGIP) Awarded by the Indian Statistical Institute and IUPRAI for best project at the Computer Vision Summer School	2018
• JBNSTS Senior Scholar (Jagadis Bose National Science Talent Search) One of 47 selected across Science, Engineering, and Medicine; among India's oldest, most selective science talent awards	2016
• KVPY Awardee (Kishore Vaigyanik Protsahan Yojana) All-India Rank 143/100,000+ (top 0.14%); prestigious national fellowship awarded by Govt. of India and IISc Bangalore	2015
• NTSE Scholar (National Talent Search Examination) Top 1,000 out of ~1,000,000 examinees (top 0.1%); Govt. of India national scholarship for academic excellence	2012

Invited Talks

• Invited Talk at School of Computing Science, Simon Fraser University, CANADA <i>ProofBridge: Auto-Formalization of Natural Language Proofs in Lean via Joint Embeddings</i>	Nov 2025
• Invited Talk at Mathematical AI Seminar, Fields Institute, University of Toronto, CANADA <i>AI for Math: Neuro-Symbolic Auto-Formalization into Lean via Joint Embeddings</i> [Abstract] [YouTube]	Oct 2025
• Technical Track Talk at 25th High Confidence Software & Systems Conference (HCSS), USA <i>Neuro-Symbolic Techniques for LLM-based Code Generation and Auto-Formalization of Proofs</i> [Slides] [Proceedings]	May 2025
• Invited Talk at ACM Student Chapter, Heritage Institute of Technology, Kolkata, INDIA <i>Neuro-Symbolic Approaches for Fine-Tuning Code Language Models</i> [Flyer]	Jan 2025
• Invited Talk at Jagadis Bose Scholars Professional Development Forum <i>Counterfactual Fairness in Machine Learning</i> [YouTube] [Flyer]	May 2021

Teaching Experience

• Co-Organizer & Speaker, PLSE/FM + AI Workshop [Site], Georgia Tech · Google · Oak Ridge National Lab · GTRI Delivered a talk on Neuro-Symbolic AI for Code & Math to 150+ PhD students, faculty, and industry professionals.	Apr 2026
• Co-Organizer & Instructor, IDEaS One-Day Tutorial on AI4Science [Tutorial Site], Georgia Tech, USA Delivered lectures and hands-on sessions on LLMs and formal methods to 80+ Ph.D. students, postdocs, and scientists.	Oct 2025
• Teaching Assistant, Georgia Tech, USA CS 3600: <i>Introduction to Artificial Intelligence</i> (Mr. Sourish Dhekane; 120+ students)	Summer 2024
• Teaching Assistant, University of Waterloo, CANADA CS 240: <i>Data Structures and Data Management</i> (Prof. Éric Schost; 400+ students) CS 105: <i>Introduction to Computer Programming</i> (Dr. Muddassir Malik; 60+ students) CS 135: <i>Designing Functional Programs</i> (Dr. Byron Weber Becker; 1,200+ students)	Fall 2022 · Spring 2023 · Summer 2023
• Teaching Assistant, IIT Kharagpur, INDIA CS 19001: <i>Programming & Data Structures Lab</i> (Drs. Abir Das & Saptarshi Ghosh; 100+ students) CS 60075: <i>Natural Language Processing</i> (Prof. Sudeshna Sarkar; 120+ students)	Fall 2021

Academic Services & Responsibilities

• IEEE Student Member Member ID: 92577001	2017 – present
• Lead Student Organizer, Foundations of AI (FoAI) Seminar Series [Seminar Page], Georgia Tech, USA Lead organizer for bi-weekly talks hosted by Profs. V. Ganesh, D. Sherrill, and C. Lin; single-handedly manage all logistics: coordinating with visiting faculty and industry researchers, venue, catering, and announcements.	2025 – present

Conference & Journal Services

2020 – present

PC Member: **AAAI** (2026, 2025, 2024, 2023 [AIES]) · **ICLR** (2026, 2025, 2024) · **ICML** (2026, 2025) ·
/ Reviewer **AISTATS** (2026, 2025) · **NeurIPS** (Main + D&B) (2026, 2025, 2024 [Top Reviewer], 2023, 2021) ·
LICS (2026) · **ECAI** (2025) · **CAV** (2025) · **NeSy** (2024, 2023) · **PLDI** (2024) · **SAT** (2024) ·
CP (2024) · **KDD** (2022) · **ACM TOPLAS** (2026) · **TMLR** (2026, 2025) · **Elsevier ASC** (2025) ·
Nature SR (2024) · **IEEE SMC** (2023) · **IEEE Access** (2022) · **Springer JCST** (2020)

Attended: ICLR 2026 (Brazil) · ICSE 2026 (Brazil) · HCSS 2025 (USA) · NeurIPS 2024 (Canada) ·
ECAI 2024 (Spain) · IJCAI 2024 (South Korea) · NeurIPS 2023 (USA)

Academic Research Internship Experience

Indian Institute of Technology (IIT) Kharagpur, INDIA

Apr 2021 – Aug 2022

Research Intern · Video Action Recognition & Zero-Shot Temporal Detection

- Advisors: Dr. Abir Das (IIT Kharagpur), Dr. Rameswar Panda (MIT-IBM Watson AI Lab)
- Extended R-C3D to achieve zero-shot end-to-end temporal activity detection in untrimmed videos; tackled recognition and localization of human activities under open-vocabulary label settings.

University of Alberta, CANADA

Jun – Dec, 2021

Research Intern · 3D Object Detection & Pose Estimation in Indoor Scenes

- Advisor: Prof. Nilanjan Ray (University of Alberta)
- Detected and estimated poses of 3D objects in indoor scenes from RGB-D images and point clouds using PointNet and Transformer architectures.

Indian Statistical Institute & IUPRAI, INDIA

May – Jul, 2018 & 2019

Research Project Trainee & Summer Intern · Spatio-Temporal Event/Activity Recognition in Videos

- Advisor: Dr. Partha Pratim Mohanta (ISI Kolkata)
- Awarded: ‘Special-Mention’ at the 5th Summer School on CV, Graphics & Image Processing (2018)
- Proposed a hierarchical decision-fusion framework combining frame- and video-level CNN & LSTM features to capture short- and long-term motion; designed a graph-based key-frame selection algorithm.

R. C. Bose Centre for Cryptology & Security, Indian Statistical Institute, INDIA

May – Jul, 2018

Summer Research Intern · Carry-less Karatsuba Multiplication over Galois Field in Block Ciphers

- Advisor: Dr. Debrup Chakraborty (ISI Kolkata)
- Supported & funded by: Microsoft Research India · Selected among: 25 students nationwide
- Implemented carry-less Karatsuba multiplication of degree-127 binary polynomials via 64-bit operations and fast polynomial reduction, accelerating AES-GCM in high-performance symmetric-key block ciphers.

Jadavpur University, INDIA

May 2017 – Oct 2018

Undergraduate Researcher · Document Analysis & Medical Image Segmentation

- Advisors: Dr. Ram Sarkar⁽¹⁾, Profs. Mita Nasipuri⁽²⁾ & Subhadip Basu⁽³⁾ (Jadavpur University)
- Proposed background inpainting and fuzzy clustering for binarizing degraded documents^(1,2); developed binarization of 3D microscopic mouse brain images^(1,3) for dendritic spine analysis.

Academic Projects & Presentations

- **Logic in Computer Science**, Georgia Tech, USA [PDF] 2023
Implemented a DPLL SAT solver; investigated the effect of branching heuristics on compute time and recursive call count
- **Advanced Topics in Requirements Engineering**, University of Waterloo, CANADA [PDF] [Slides] 2023
Designed the requirements engineering process for an LLM-based source code translation system
- **Distributed Systems**, IIT Kharagpur, INDIA [Report] [PDF] 2021
Analyzed distributed pub-sub event streaming performance of Apache Pulsar, Google Cloud Pub/Sub, and Amazon Kinesis
- **High Performance Computer Architecture**, IIT Kharagpur, INDIA [Report] [PDF] 2021
Used gem5 to simulate the N-queens problem; analyzed CPI variation across L2, L1-i/d cache sizes, and branch predictors
- **Artificial Intelligence**, IIT Kharagpur, INDIA [YouTube] [Report] [PDF] 2020
Analyzed perception and route planning algorithms for autonomous vehicles, interpreting multi-modal sensor data

Relevant Coursework Pursued

[During Ph.D.] Georgia Institute of Technology, USA & University of Waterloo, CANADA

2022 – 2025

(CS 6241) Advanced Compiler Optimization · Prof. Santosh Pande

(CS 8903-SAT) SAT and SMT Solvers: Theory, Practice, and Applications · Prof. Vijay Ganesh

(CS 6340) Software Analysis and Test · Prof. Qirun Zhang

(CS 6601) Artificial Intelligence · Prof. Thomas Ploetz

(CS 8903-LCS) Logic in Computer Science · Prof. Suguman Bansal

(CS 846) Advanced Topics in Software Engineering: Requirements Engineering · Prof. Daniel Berry

(CS 898) Advanced Special Topics in Computer Science: Computer Vision · Prof. Yuri Boykov

(CS 685) Machine Learning: Statistical & Computational Foundations · Prof. Shai Ben-David

[During Master's] IIT Kharagpur, INDIA (Total Credit Hours: 49)

2020 – 2022

Algorithm Design & Analysis, AI, NLP, Foundations of CS, Info Retrieval, Complex Networks, High-Performance Computer Architecture, Computational Geometry, Deep Learning, Distributed Systems

[During Bachelor's] Jadavpur University, INDIA (Total Credit Hours: 193)

2016 – 2020

Digital Logic & Circuits, Data Structures & Algo, Design & Analysis of Algo, Numerical Methods, OOP, Microprocessors & Assembly Lang, Graph Theory & Combinatorics, Computer Organization & Architecture, Systems Programming, OS, Formal Lang & Automata, Digital Communication, Computer Graphics, DBMS, Principles of Programming Lang, Compiler Design, Computer Networks, Software Engg., AI, Internet Tech, Data Mining, VLSI, Optimization Tech & Operation Research, NLP, Soft Computing